Panel Data Analysis

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Pooling Independent Cross Sections across Time

- **Pooling Independent Cross Sections across Time**
  - Many surveys of individuals, families, and firms are repeated over time with new random samples obtained in each period. Examples include the "General Social Survey" and the "Current Population Survey".
  - Pooling the random samples gives us an independently **pooled cross section.**
  - Statistically, analyzing pooled cross sections is similar to a single cross section, provided we assume random sampling was used to collect each cross section
  - One reason for using pooled cross sections is to increase the sample size
  - With a pooled cross section, often a **goal is to see how the mean value of a variable has changed over time** in ways that cannot be explained by observable variables

Pooling Independent Cross Sections across Time

- To reflect the fact that the population may have different distributions in different time periods, we can **include dummy variables for each period**.
- **Example: Women's Fertility over Time** (FERTIL1.dta)
  - The data set comes from the "General Social Survey" for the even years from 1972 to 1984.
  - *After* controlling for education, has the pattern of fertility among women over age 35 changed between the sample periods?

$$kids = \beta_0 + \beta_1 educ + ... + \delta_0 y74 + \delta_1 y76 + ... + \delta_5 y84 + u$$

Total number of kids born to a women

Dummy variables for all but one year, where 1972 is chosen as the base year

  - This is call **pooled OLS regression,** or **POLS regression**

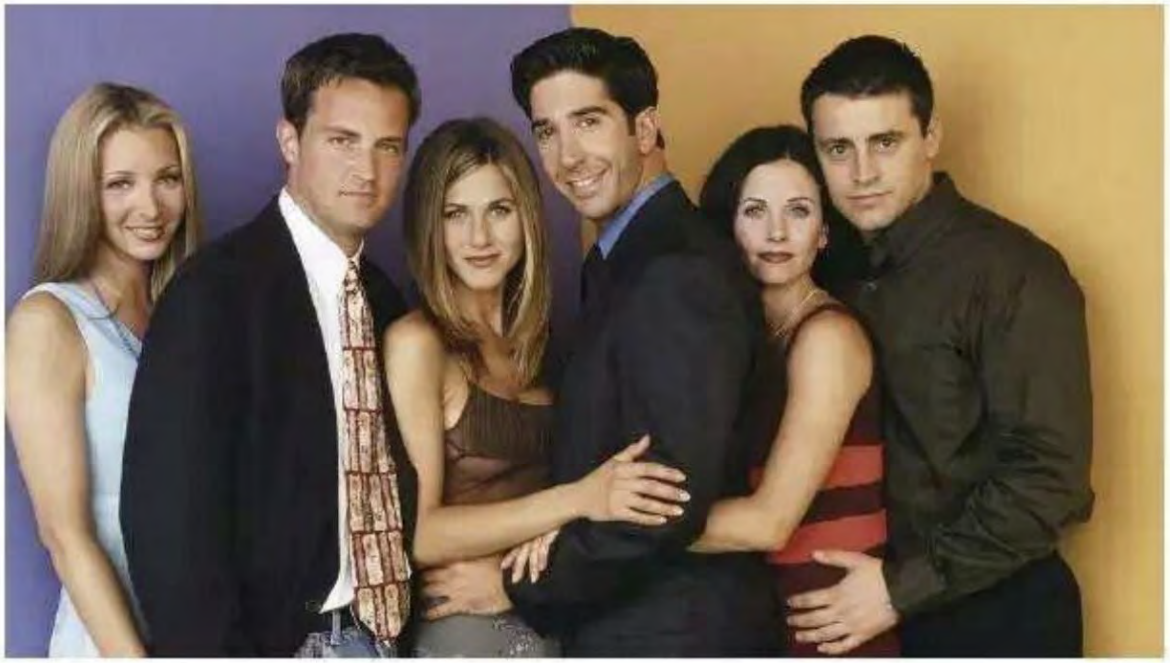
What Are Panel Data?

A panel dataset contains observations for multiple entities (individuals, states, etc.), where each entity is observed at two or more time periods (months, years, etc.). Also called longitudinal data.

Examples:

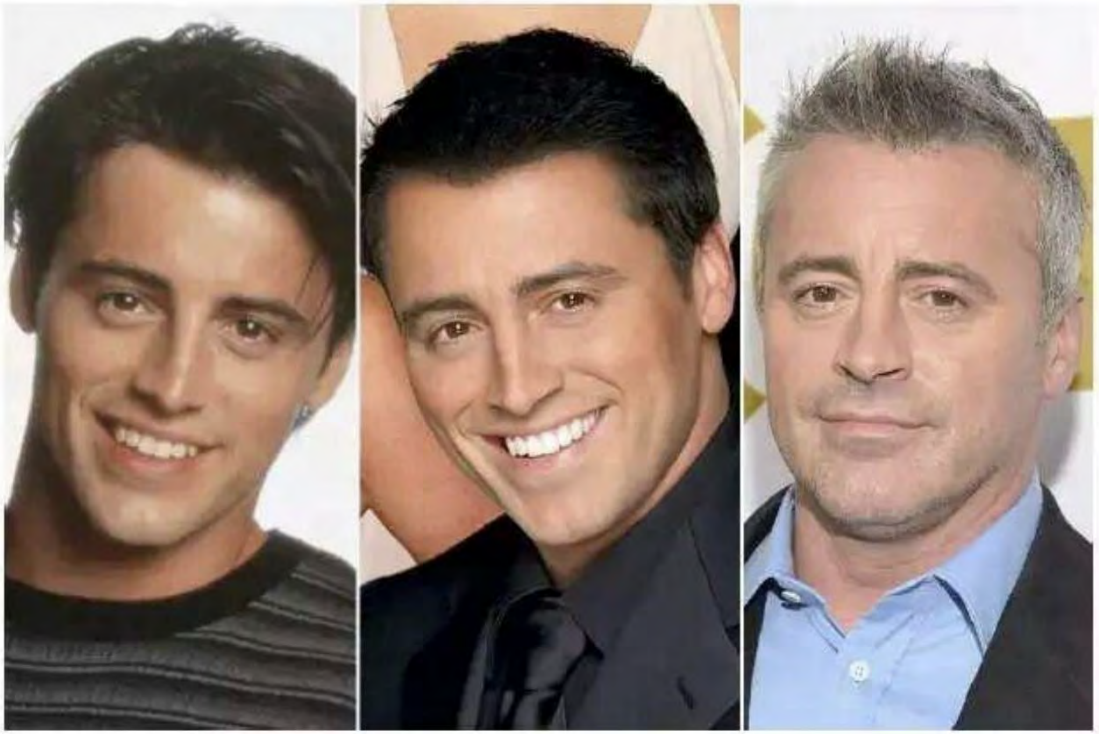
- Data on 420 California school districts in 1999 and in 2000, for a total of 840 observations
- Data on 50 U.S. states, where each state is observed in 3 years, for a total of 150 observations
- Data on 1000 individuals, in four different months, for a total of 4000 observations

The structure of economic data



2003

The structure of economic data



1993 2003 2013

The structure of economic data



1993



2003



2013

The structure of economic data



Notations for Panel Data

A double subscript is used to distinguish entities (individuals, states, etc.) and time periods (months, years, etc.):

- i = entity, n = the total number of entities, so $i = 1, \ldots, n$
- t = time period, T = the total number of time periods, so $t = 1, \ldots, T$

Data: Suppose we have k regressors. The data are:

$$(X_{1it}, X_{2it}, \ldots, X_{kit}, Y_{it}), i = 1, \ldots, n, t = 1, \ldots, T .$$

- Balanced panel: no missing observations
- Unbalanced panel: some entities (individuals, states, etc.) are not observed for some time periods (months, years, etc.)

Why Are Panel Data Useful?

With panel data we can control for factors that:

- Are unobserved or unmeasured – and therefore cannot be included in the regression using multiple regression
- Could cause omitted variable bias if they are omitted
- Vary across entities but do not vary over time

Here's the key idea:

If an omitted variable does not change over time, then any changes in Y over time cannot be caused by the omitted variable – the omitted variable can be removed by some methods.

Example of A Panel Data Set

Example: Traffic deaths and alcohol taxes

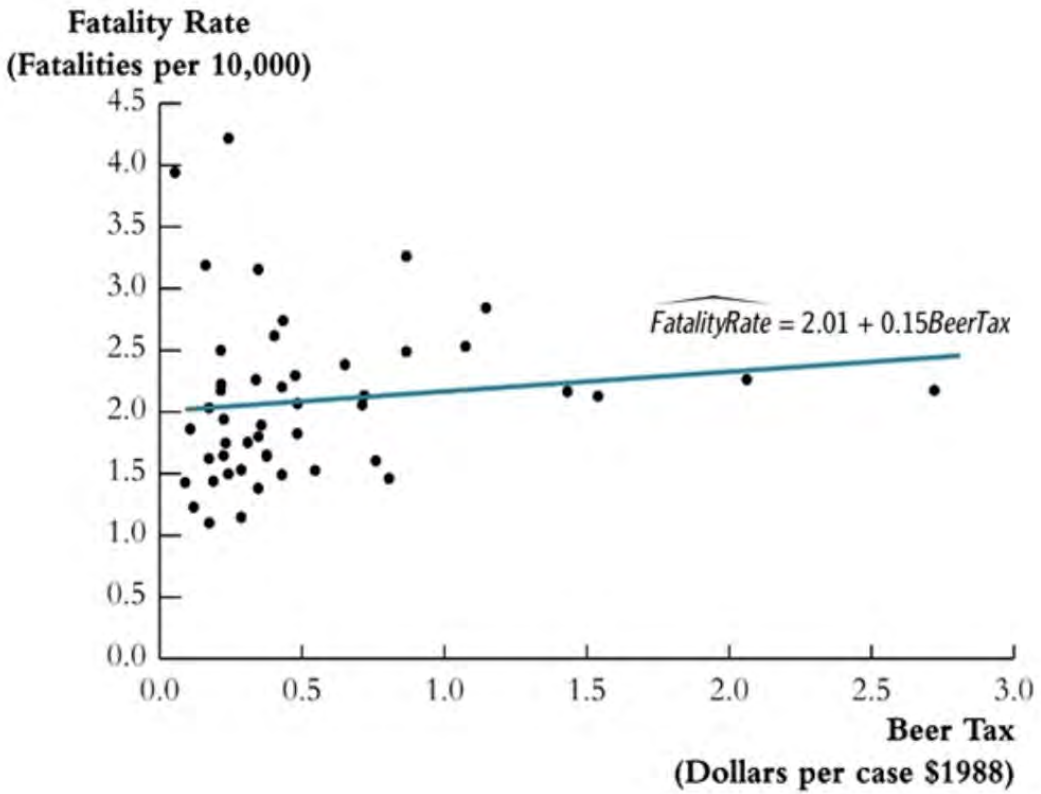
Observational unit: a year in a U.S. state

- 48 U.S. states, so n = # of entities = 48
- 7 years (1982,...,1988), so T = # of time periods = 7
- Balanced panel, so total # of observations = $7 \times 48 = 336$

Variables:

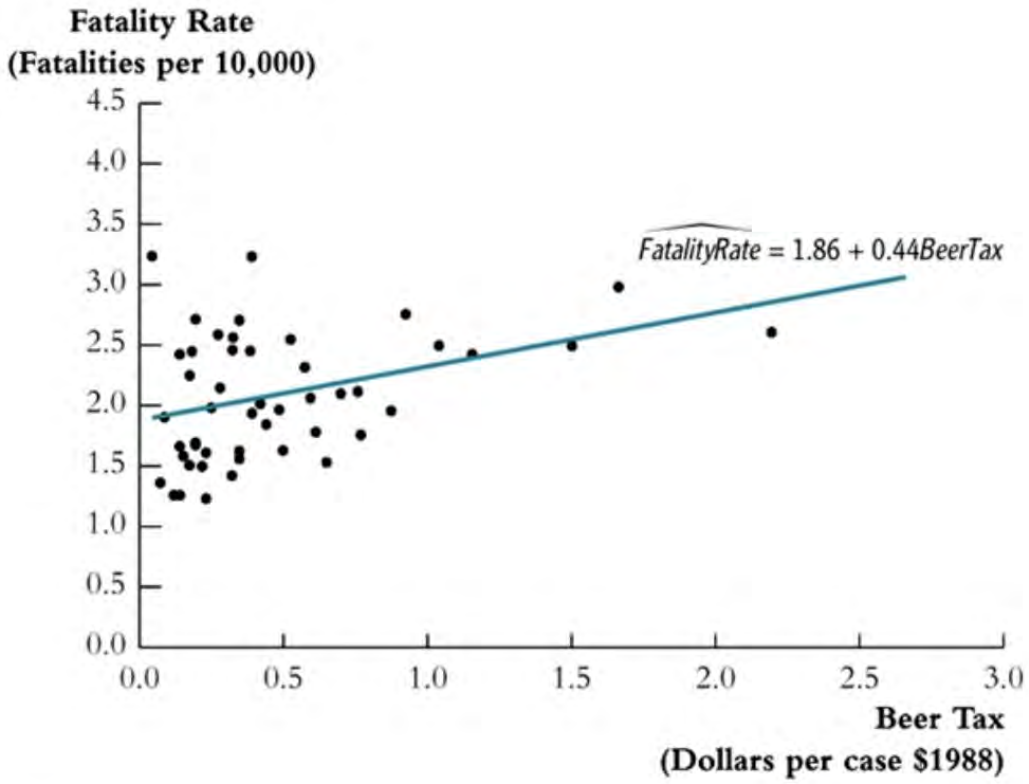
- Traffic fatality rate (# of traffic deaths in that state in that year, per 10,000 state residents)
- Tax on a case of beer
- Others (legal driving age, drunk driving laws, etc.)

Traffic Death Data for 1982



Higher alcohol taxes, more traffic deaths?

Traffic Death Data for 1988



Higher alcohol taxes, more traffic deaths?

Omitted Variable Bias

Why might there be more traffic deaths in states that have higher alcohol taxes?

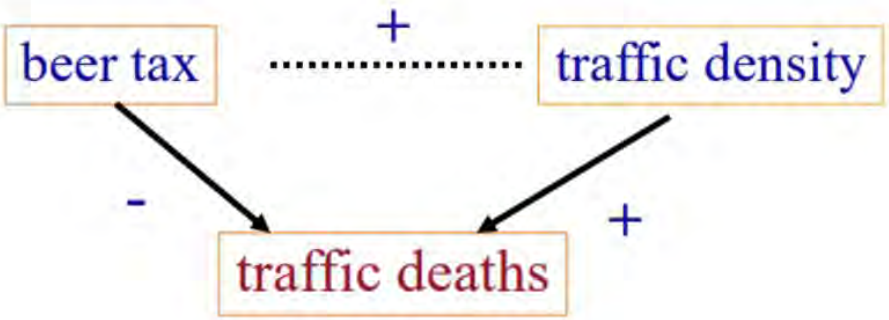
Other factors that determine traffic fatality rate:

- Quality (age) of automobiles
- Quality of roads
- "Culture" around drinking and driving
- Density of cars on the road

Omitted Variable Bias

Example: Traffic density

- High traffic density means more traffic deaths
- States with lower traffic density have lower alcohol taxes



$$\text{FatalityRate}_{it} = \beta_0 + \beta_1 \text{BeerTax}_{it} + \gamma \text{TrafficDensity}_{it} + e_{it} ,$$

$\gamma > 0$ and $corr(\text{BeerTax}, \text{TrafficDensity}) > 0 \Rightarrow$ Positive bias

Basic Idea of Using Panel Data

- Panel data lets us eliminate omitted variable bias when the omitted variable is constant over time within a given state
- Intuitively, if an omitted variable is constant over time, then any changes in Y over time cannot be caused by the omitted variable
- Mathematically,if an omitted variable does not change over time, we can remove the omitted variable by taking the first difference of the data across time periods

Panel Data with T = 2

Consider the panel data model

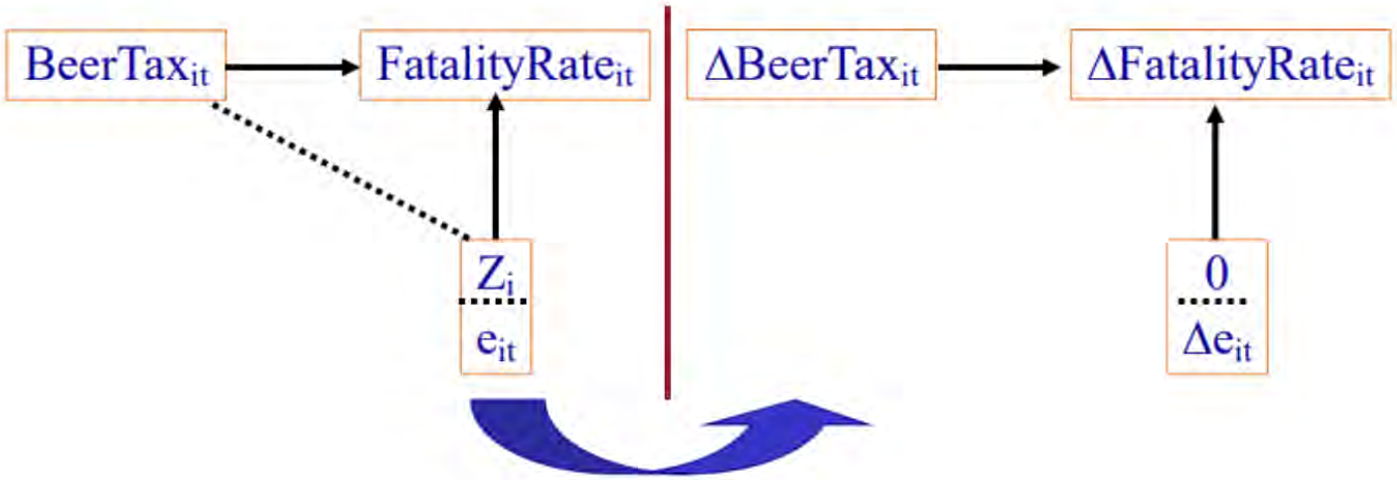
$$\text{FatalityRate}_{it} = \beta_0 + \beta_1 \text{BeerTax}_{it} + \gamma Z_i + e_{it} \ ,$$

where Z_i is a factor that does not change over time, at least during the years on which we have data. (It is reasonable to assume traffic density does not change between two years)

- Suppose Z_i is not observed, so its omission could result in omitted variable bias
- The omitted variable bias can be eliminated using $T = 2$ years

Panel Data with T = 2

The key idea: Any change in the fatality rate from 1982 to 1988 cannot be caused by Z_i, because Z_i (by assumption) does not change between 1982 and 1988.



The omitted variable Z_i doesn't change, so it cannot be a determinant of the change in Y

Panel Data with T = 2

Mathematically,

- When $t = 1982$,

$$\text{FatalityRate}_{i1982} = \beta_0 + \beta_1 \text{BeerTax}_{i1982} + \gamma Z_i + e_{i1982} \,,$$

- When $t = 1988$,

$$\text{FatalityRate}_{i1988} = \beta_0 + \beta_1 \text{BeerTax}_{i1988} + \gamma Z_i + e_{i1988} \,,$$

- Take the difference,

$$\begin{aligned} &\text{FatalityRate}_{i1988} - \text{FatalityRate}_{i1982} \\ &\quad = \beta_1(\text{BeerTax}_{i1988} - \text{BeerTax}_{i1982}) + (e_{i1988} - e_{i1982}) \,. \end{aligned}$$

The omitted variable Z_i has been successfully removed!

Panel Data with T = 2

The difference equation:

$$\begin{aligned}&\text{FatalityRate}_{i1988} - \text{FatalityRate}_{i1982} \\ &\quad = \beta_1(\text{BeerTax}_{i1988} - \text{BeerTax}_{i1982}) + (e_{i1988} - e_{i1982}) .\end{aligned}$$

Suppose $E[e_{it}|\text{BeerTax}_{it}] = 0$. Then

- The new error term, $e_{i1988} - e_{i1982}$, is uncorrelated with either BeerTax_{i1988} or BeerTax_{i1982}
- This "difference" equation can be estimated by OLS, even though Z_i isn't observed. That is, the causal effect β_1 can be estimated by regressing $\text{FatalityRate}_{i1988} - \text{FatalityRate}_{i1982}$ on $\text{BeerTax}_{i1988} - \text{BeerTax}_{i1982}$

Panel Data with T = 2

Mathematically,

- For $t = 1982$,

$$\widehat{\text{FatalityRate}}_{i1982} = \underset{(0.15)}{2.01} + \underset{(0.13)}{0.15} \times \text{BeerTax}_{i1982}, n = 48$$

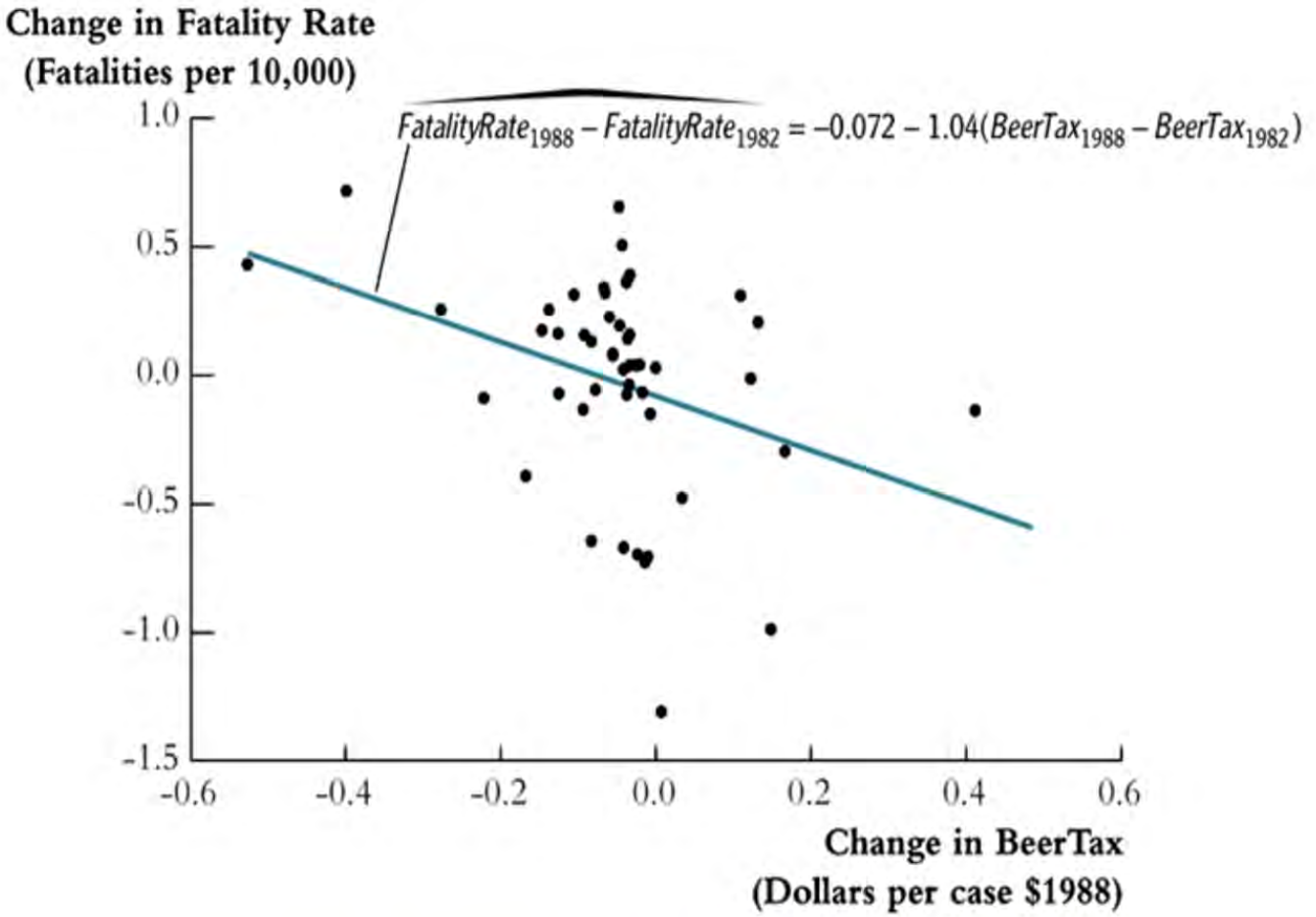
- For $t = 1988$,

$$\widehat{\text{FatalityRate}}_{i1988} = \underset{(0.11)}{1.86} + \underset{(0.13)}{0.44} \times \text{BeerTax}_{i1988}, n = 48$$

- For the difference equation,

$$\widehat{\text{FatalityRate}_{i1988} - \text{FatalityRate}_{i1982}} = \underset{(0.065)}{-0.072} - \underset{(0.36)}{1.04} \times (\text{BeerTax}_{i1988} - \text{BeerTax}_{i1982})$$

Panel Data with T = 2



Higher alcohol taxes, less traffic deaths!

Panel Data with T > 2

What if you have more than 2 time periods ($T > 2$)?

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \gamma Z_i + e_{it}, i = 1, \ldots, n, t = 1, \ldots, T .$$

We can rewrite this in two useful ways:

1. "Fixed effects" regression model
2. "$n - 1$ binary regressor" regression model

Fixed Effects Form

We first rewrite the model in "fixed effects" form. Suppose we have $n = 3$ states: California, Texas, New York.

Population regression for California (that is, $i = CA$):

$$\begin{aligned} Y_{CA,t} &= (\beta_0 + \gamma Z_{CA}) + \beta_1 X_{CA,t} + e_{CA,t} \\ &= \alpha_{CA} + \beta_1 X_{CA,t} + e_{CA,t} \, . \end{aligned}$$

- $\alpha_{CA} = \beta_0 + \gamma Z_{CA}$ doesn't change over time
- α_{CA} is the intercept for CA, and β_1 is the slope
- The intercept is unique to CA, but the slope is the same for all states: parallel lines

Fixed Effects Form

For TX and NY,

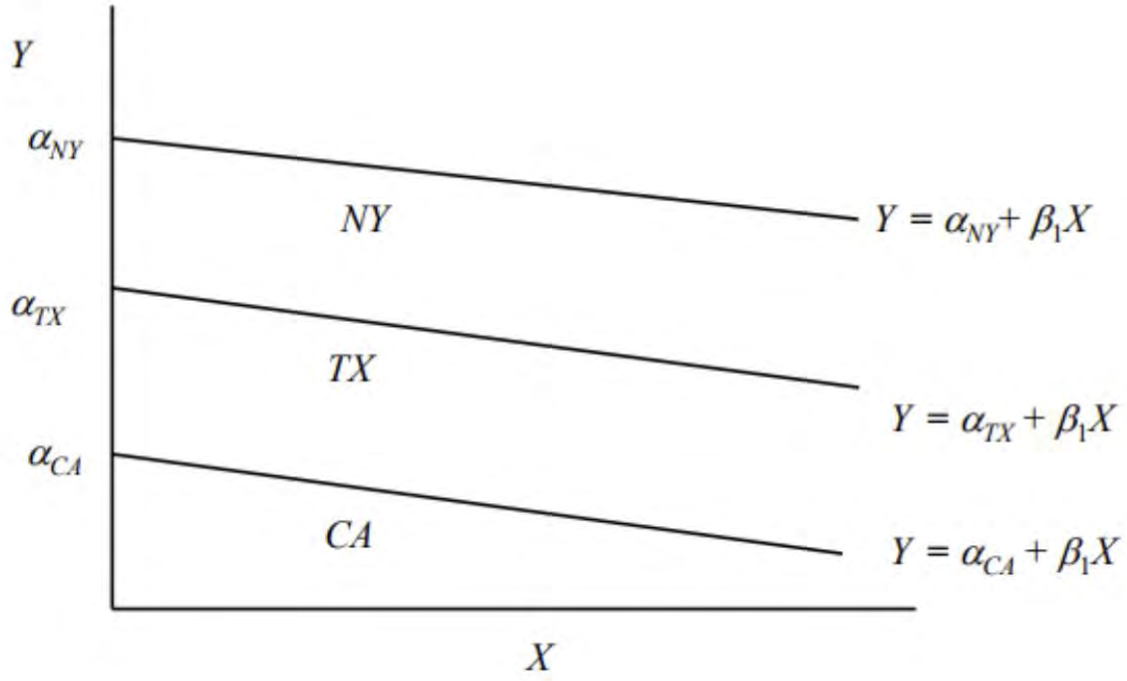
$$Y_{TX,t} = \alpha_{TX} + \beta_1 X_{TX,t} + e_{TX,t}$$
$$Y_{NY,t} = \alpha_{NY} + \beta_1 X_{NY,t} + e_{NY,t} .$$

Collecting the lines for all three states:

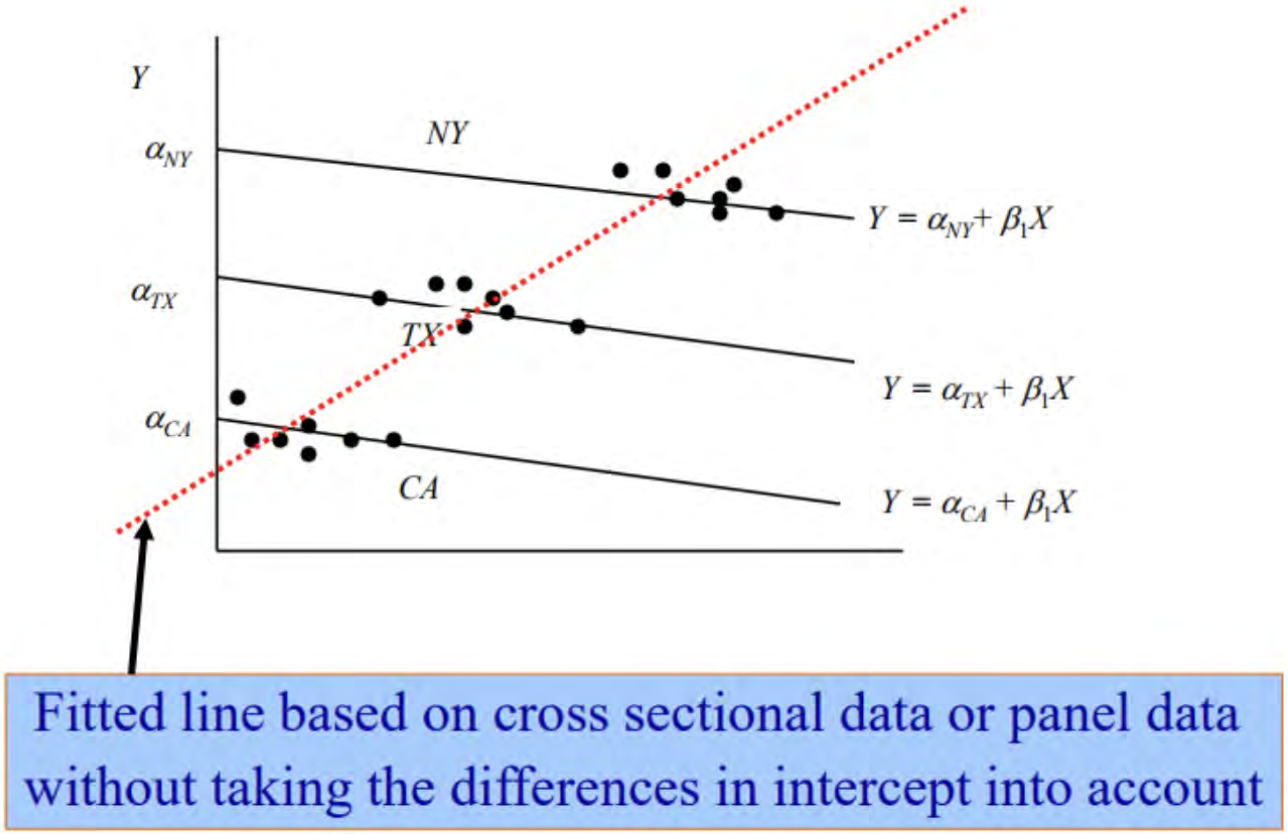
$$Y_{it} = \alpha_i + \beta_1 X_{it} + e_{it} \ , i = CA, TX, NY, t = 1, \ldots, T .$$

- Three parallel regression lines with different intercepts

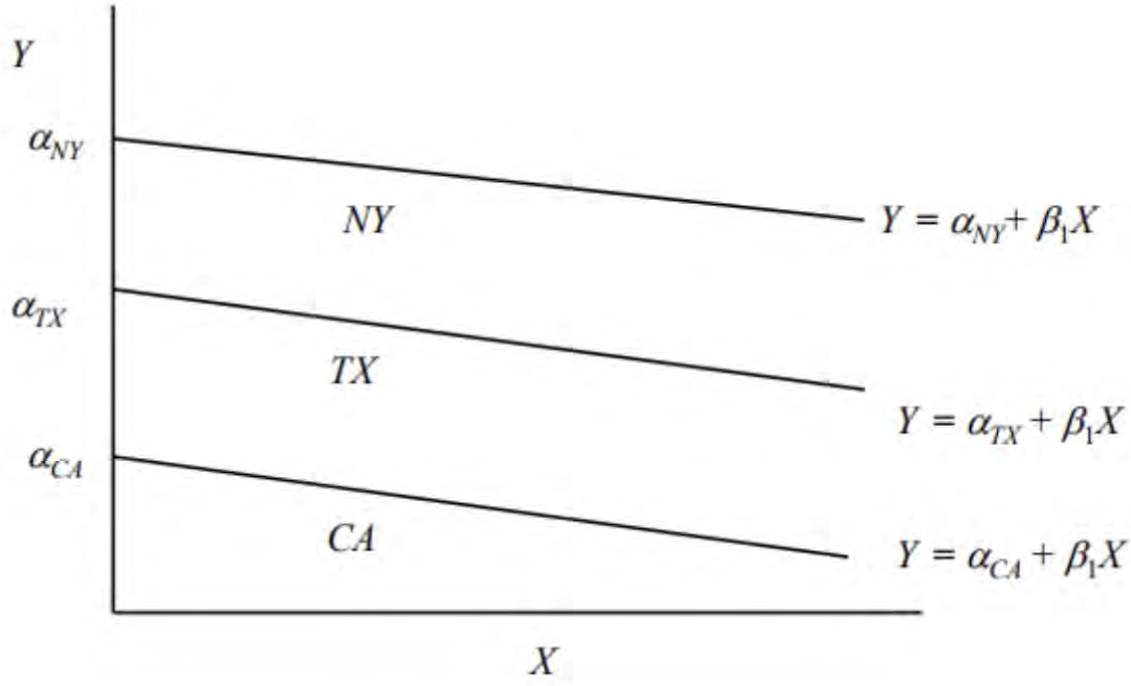
Fixed Effects Form



Fixed Effects Form

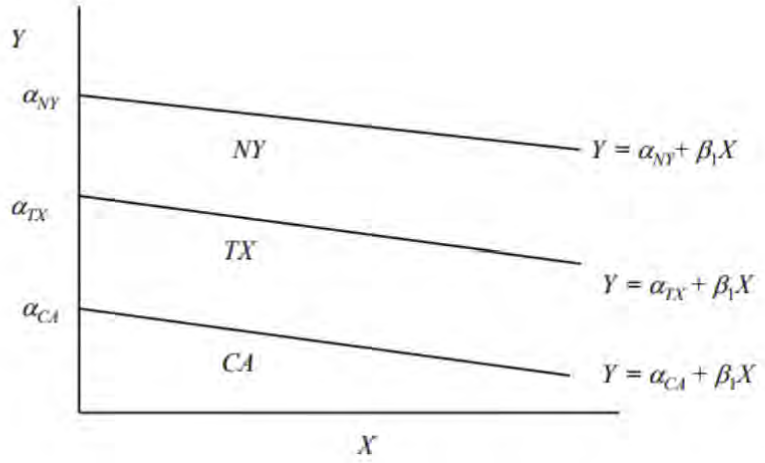


Binary Regressor Form



Recall that shifts in the intercept can be represented using binary regressors (dummy variables)

Binary Regressor Form



In binary (dummy) regressor form:

$$Y_{it} = \beta_0 + \gamma_{CA} DCA_i + \gamma_{TX} DTX_i + \beta_1 X_{it} + e_{it}$$

- $DCA_i = 1$ if state is CA, and 0 otherwise
- $DTX_i = 1$ if state is TX, and 0 otherwise
- Leave out DNY_i (why?)

Binary Regressor Form

$$Y_{it} = \beta_0 + \gamma_{CA} DCA_i + \gamma_{TX} DTX_i + \beta_1 X_{it} + e_{it}$$

	Constant	DCA_i	DTX_i	DNY_i
i=CA:	1	1	0	0
i=TX:	1	0	1	0
i=NY:	1	0	0	1

$$\text{Constant} = DCA_i + DTX_i + DNY_i$$

- Multicollinearity between the constant regressor and the dummy variables
- Solution : choose a base group, say DNY_i and drop it from the regression
- Intercept for NY : β_0; Intercept for CA : $\beta_0 + \gamma_{CA}$; Intercept for TX : $\beta_0 + \gamma_{TX}$

Summary: Two Forms

Fixed effects form:

$$Y_{it} = \beta_1 X_{it} + \alpha_i + e_{it} \ ,$$

where α_i is called a "fixed effect" or "entity fixed effect" or "entity effect" – it is the constant (fixed) effect of being entity i.

Binary regressor form:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \gamma_2 D2_i + \cdots + \gamma_n Dn_i + e_{it} \ ,$$

where $D2_i = 1$ for $i = 2$ (entity 2) and 0 otherwise, etc.